SWT 22022 – Practical for Internet Application Development

Department of Information and Communication Technology

Faculty of Technology

Lab sheet :17
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Title: Introduction to React and Basic Functional Components

Aims:

- Setting up development environment
- Project creation
- Understanding the core concepts

Tasks:

- Install Node.js and npm
- Setting up React
- Start the development server
- Basic React Components

Activities :

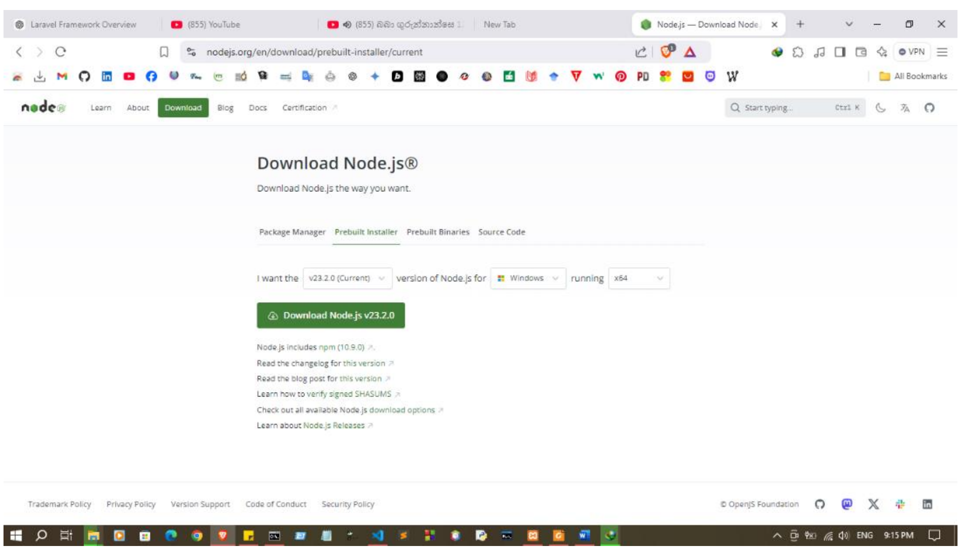
1. Install Node.js and npm

Windows:

- Download the appropriate Node.js installer from the official website:

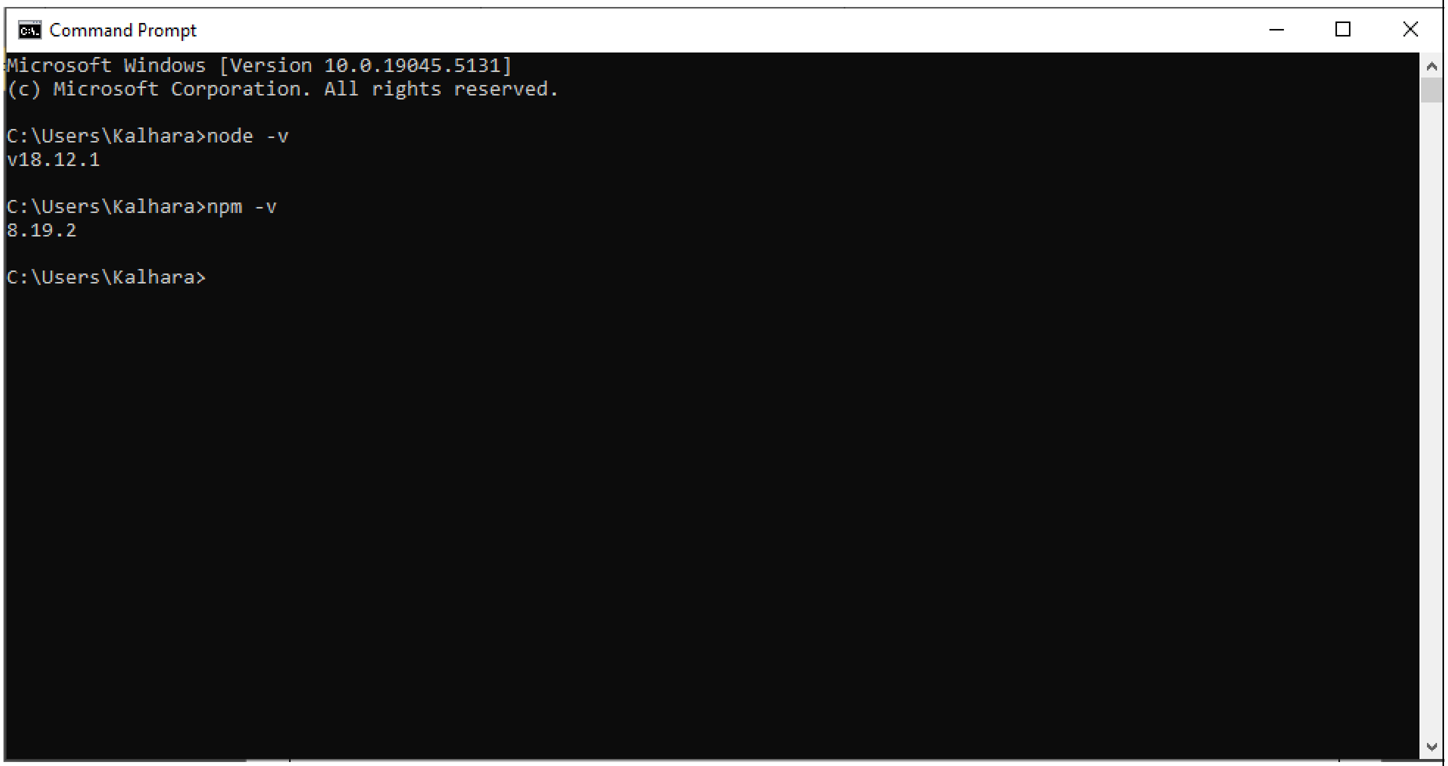
  https://node.js.org/en/download/

- Run the downloaded installer file and follow the on-screen instructions.



- Verify the installation by opening a command prompt and typing

  node -v and npm -v



macOS:

- Install Homebrew if you don't already have it: htps://docs.brew.sh/Installation
- Once Homebrew is installed, run the following command to install Node.js:

  brew install node
- Verify the installation by opening a Terminal window and typing

  node -v and npm -v

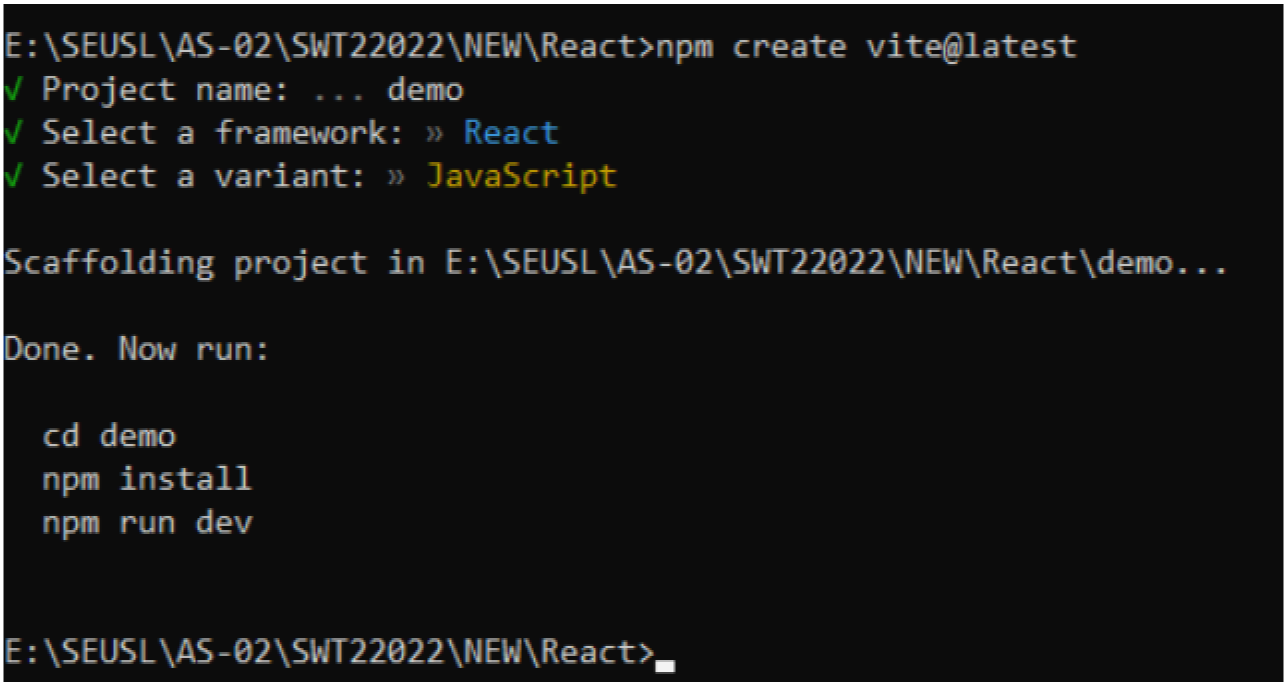
Linux:

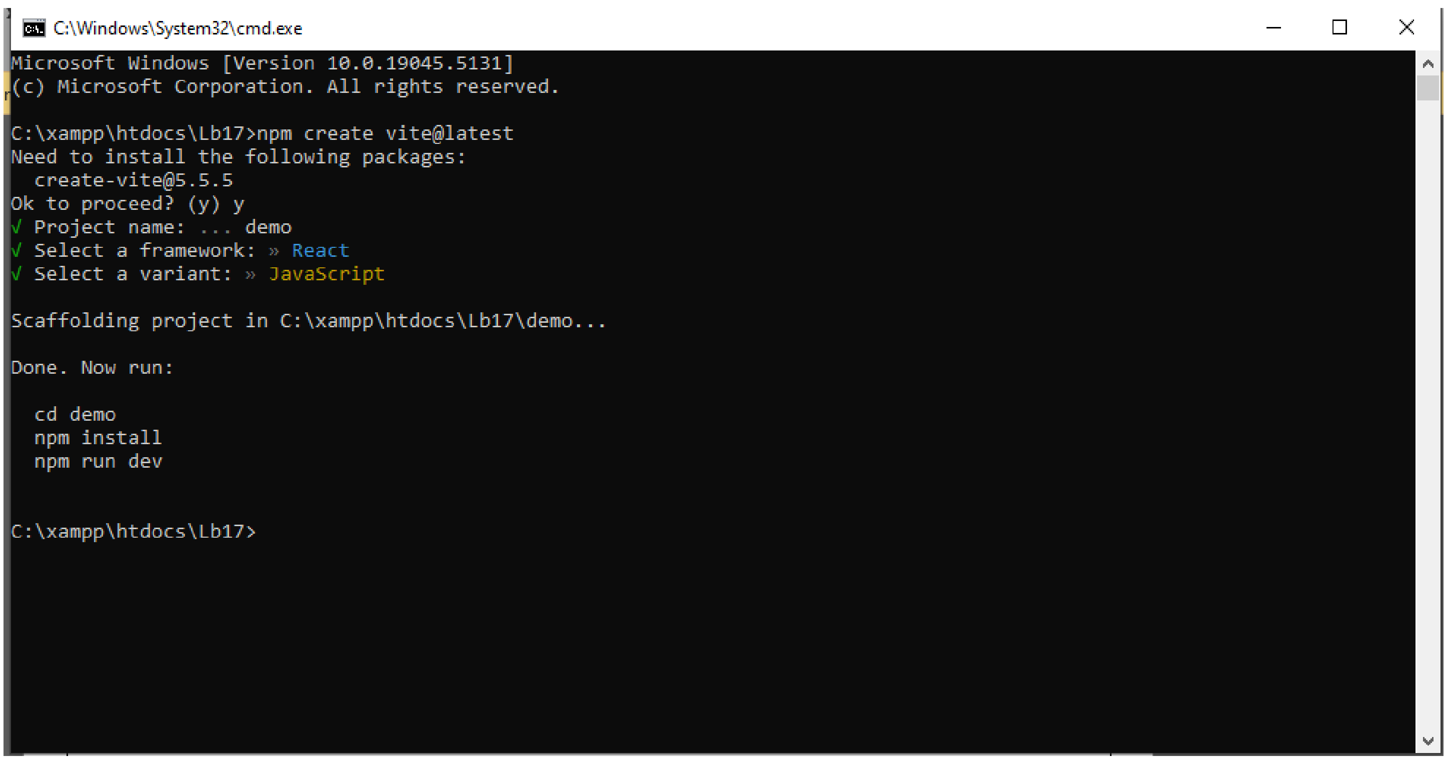
- The installation method for Linux depends on your specific distribution. However, generally, you can use your package manager to install Node.js. For example, on Ubuntu, you can use the following command:

  sudo apt install nodejs npm
- Verify the installation by opening a terminal window and typing

  node -v and npm -v

2. Setting up React



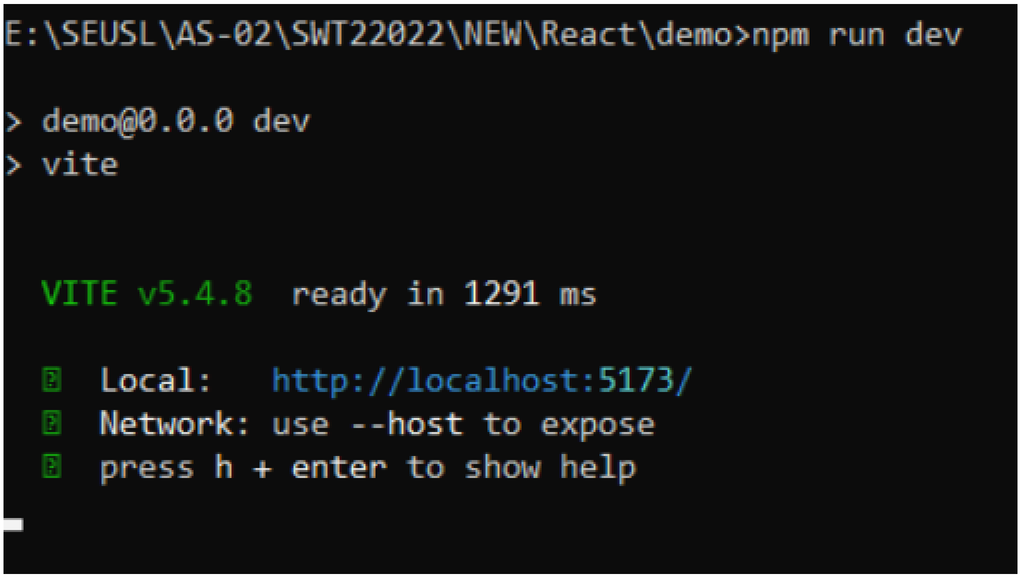


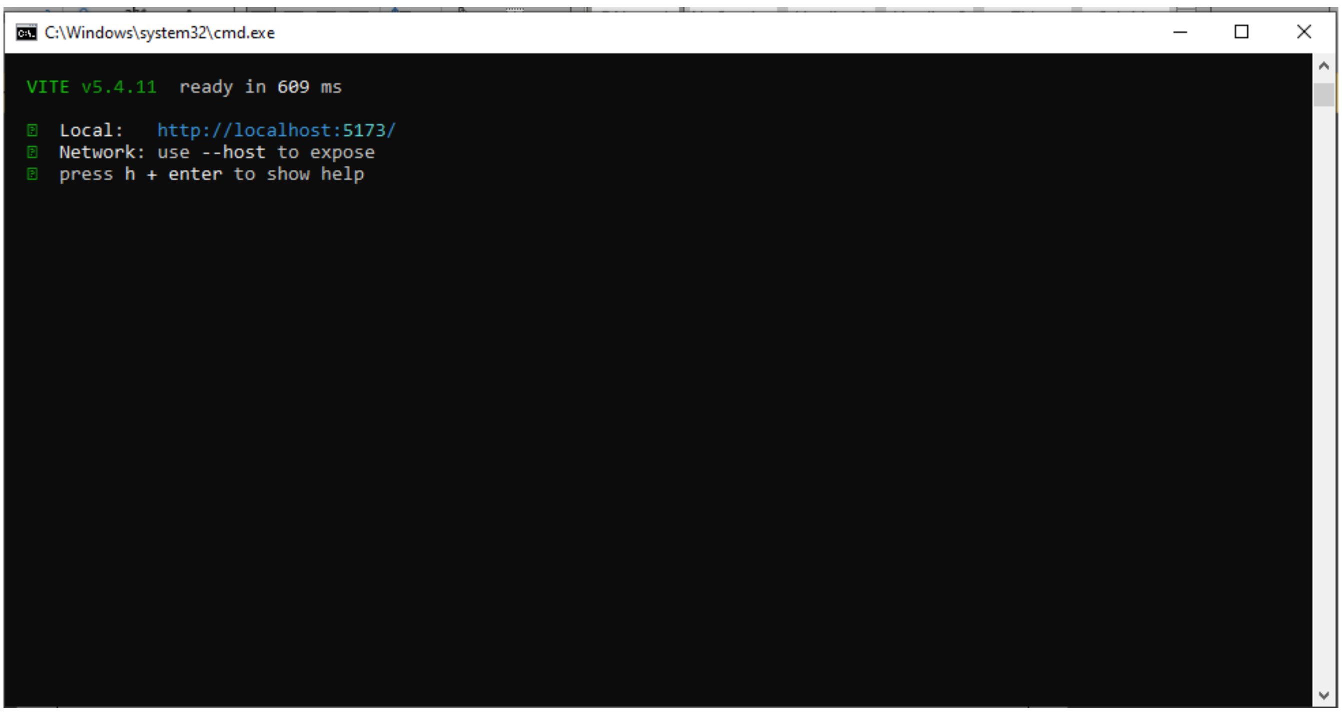
3. Start the development server

Install the required dependencies for the project:

npm install

Run the development server:





4. Basic React Components

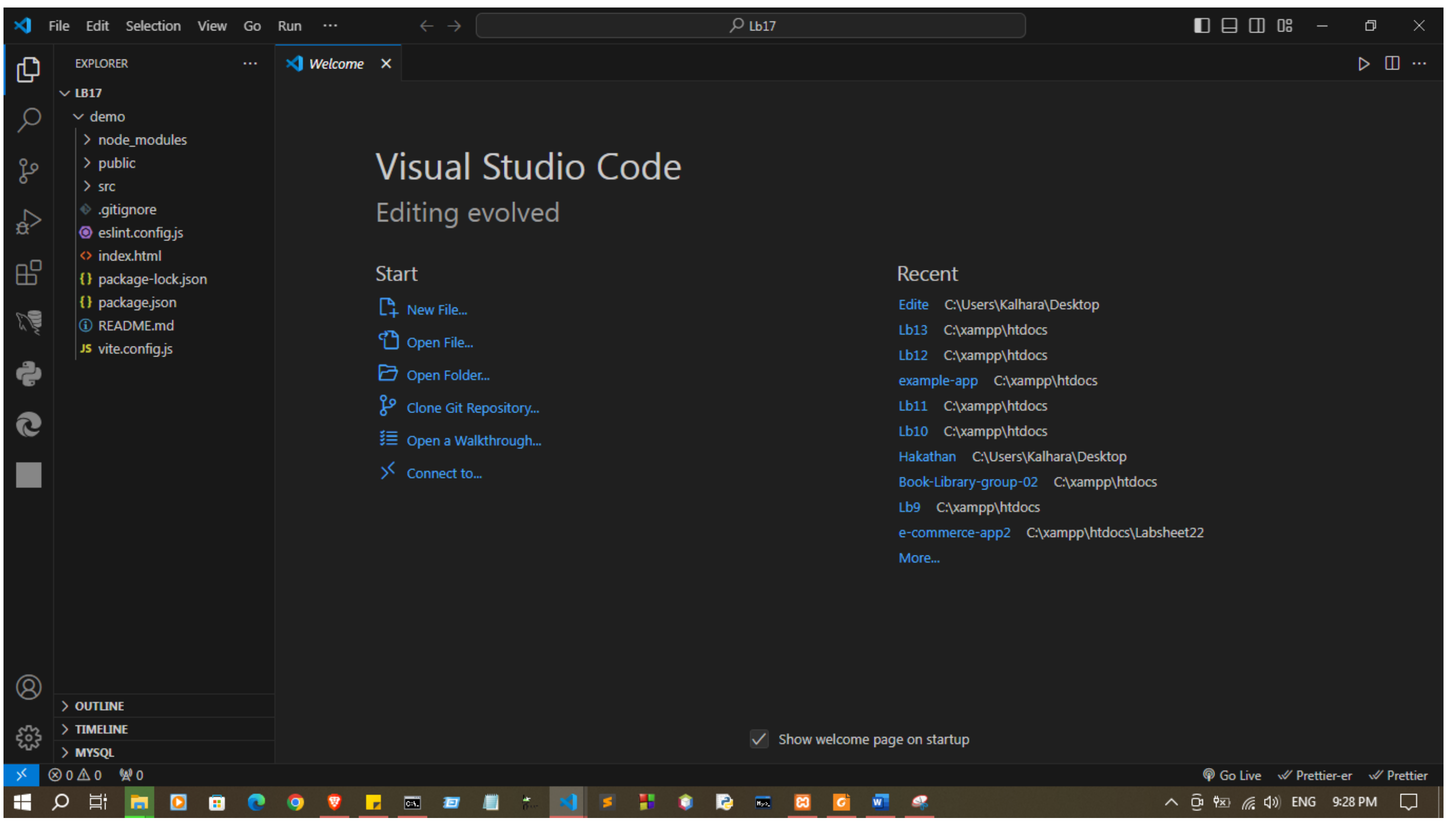
Hello.jsx

```
function Hello() {
    return (
    <>
    <div>
    <h2>Hello World !</h2>
    </div>
```

```
        </>
        )
    }
    export default Hello
```

<u>*main.jsx*</u>

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
// import App from './App.jsx'
import './index.css'
import Hello from './hello.jsx'
createRoot(document.getElementById('root')).render(
    <StrictMode>
        <Hello />
    </StrictMode>,
)
```



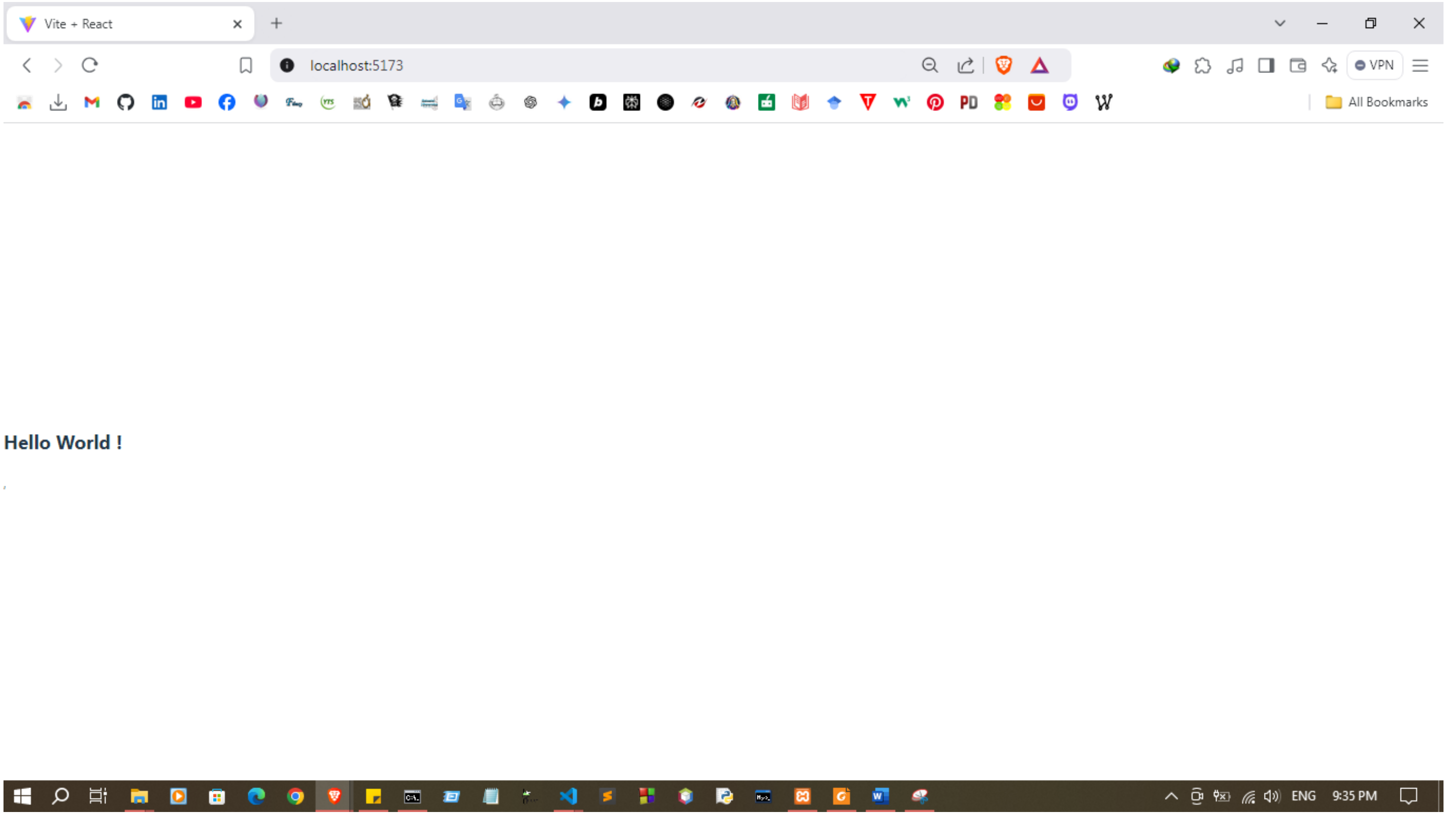
Hello.jsx

```
function Hello() {
    return (
    <>
    <div>
    <h2>Hello World !</h2>
    </div>
    </>
    )
}
export default Hello
```

App.jsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
// import App from './App.jsx'
import './index.css'
import Hello from './Hello.jsx'
function App() {
  return (
    <>
      <StrictMode>
        <Hello />
    </StrictMode>,

    </>
  )
}
export default App
```



Discussion:

- This lab session introduces React, a popular JavaScript library for building user interfaces, with a focus on basic functional components. It starts with setting up the development environment by installing Node.js and npm, which are essential tools for running and managing React projects. Participants will then learn to set up a React application and start the development server to view real-time updates as they build. The session emphasizes understanding React's core concepts, such as components, props, and the virtual DOM, while creating and working with basic functional components. By the end of this session, students will have a foundational understanding of how to structure and develop a React project effectively.